Kai Zheng | Curriculum Vitae

6085 Water Street #2360 – Plano, Tx 75024
+1 (646) 717 4078 • kaizheng28@gmail.com • kaizheng.me

Research Interest

Wireless Sensing and Networking: mmWave Radar Systems (Joint radar-communication, metasurface-assisted sensing, AI-enhanced perception); novel radio hardware and software design (Software-Defined Radios, mmWave/FR3 massive-MIMO platform).
Mobile and Ubiquitous Computing: Low-power reliable IoT networking, neuromorphic computing and sensing.

Education

University of California San Diego (UCSD) — **La Jolla, CA**
Ph.D. in Electrical and Computer Engineering — *Sept. 2020 – Sept. 2025*
Thesis: Intelligent IoT Radar Sensing with Novel Array-Centric Hardware Design
Advisor: Prof. Xinyu Zhang

New York University (NYU) — **Brooklyn, NY**
M.S. in Computer Engineering — *Sept. 2017 – May 2019*
Advisor: Prof. Sundeep Rangan

Fudan University (FDU) — **Shanghai, China**
B.S. in Electrical Engineering — *Sept. 2011 – Jul. 2015*

Work Experience

Samsung Research America — **Plano, TX, USA**
Research Engineer, Senior — *Sept. 2025 – Present*

Pi Radio Inc. — **Brooklyn, NY, USA**
Radio-Frequency Engineer — *Aug. 2019 – Aug. 2020*

Huawei Technology Co., Ltd. — **Shanghai, China**
Baseband Hardware Engineer — *Aug. 2015 – Aug. 2017*

Publication

Conference Papers

- "MetaCapsule: Toward In-Vivo Wi-Fi Sensing with a Smart Capsule"
  Baicheng Chen, **Kai Zheng**, and Xinyu Zhang.
  The ACM Conference on Embedded Networked Sensor Systems **(SenSys)**, To Appear, 2027.
  (17.7% acceptance rate, 43 of 243)
- "AutoBLE: Automating BLE Ranging Algorithm Design with Agentic Digital-Twin Optimization"
  Kai Zheng, Vishnu V. Ratnam, Priyabrata Parida, Hao Chen, Hosik Cho, Hyun-Cheol Park, and Sangyoun Lee.
  Proceedings of the ACM Workshop on Digital Twins over NextG Wireless Networks **(DTwin)**, 2026.
- "MetaVenom: Field Programmable Fully Passive Metasurfaces"
  Baicheng Chen, Justin Chen, Wuqiong Zhao, **Kai Zheng**, and Xinyu Zhang.
  The ACM International Conference on Mobile Computing and Networking **(MobiCom)**, 2026.
- "FlowForm: Scalable Passive Metasurface Network for mmWave Coverage Expansion"
  Wuqiong Zhao, Baicheng Chen, **Kai Zheng**, Xingyu Chen, Wenqian Zhang, and Xinyu Zhang.

The ACM SIGCOMM Conference **(SIGCOMM)**, 2026.

- "Physically Accurate Differentiable Inverse Rendering for Radio Frequency Digital Twin"
  Xingyu Chen, Xinyu Zhang, **Kai Zheng**, Xinmin Fang, Tzu-Mao Li, Chris Xiaoxuan Lu, and Zhengxiong Li.
  The ACM International Conference on Mobile Computing and Networking **(MobiCom)**, 2026.
  (11.2% acceptance rate)
- "MetaPolar: A Low-Cost, Passive, mmWave Radar-Readable Metasurface Road Sign"
  Kai Zheng, Wuqiong Zhao, Xinyu Zhang. The 24th ACM Conference on Embedded Networked Sensor Systems **(SenSys)**, 2026.
  (19.0% acceptance rate)
- "NEAR: Neural Electromagnetic Array Response"
  Yinyan Bu, Jiajie Yu, **Kai Zheng**, Xinyu Zhang, Piya Pal.
  Forty-Second International Conference on Machine Learning. **(ICML)**, 2025.
  (26.9% acceptance rate)
- "Enhancing mmWave Radar Sensing Using a Phased-MIMO Architecture"
  Kai Zheng, Wuqiong Zhao, Timothy Woodford, Renjie Zhao, Xinyu Zhang, Yingbo Hua.
  The 22nd Annual International Conference on Mobile Systems, Applications and Services **(MobiSys)**, 2024.
  (16.3% acceptance rate)
- "NeuroRadar: A Neuromorphic Radar Sensor for Low-Power IoT Systems"
  Kai Zheng, Kun Qian, Timothy Woodford, Xinyu Zhang.
  The 21st ACM Conference on Embedded Networked Sensor Systems **(SenSys)**, 2023.
  (19.0% acceptance rate, **Best Paper Award, Communications of the ACM Research Highlight**)
- "SlimWiFi: Ultra-Low-Power IoT Radio Architecture Enabled by Asymmetric Communication"
  Renjie Zhao, Kejia Wang, **Kai Zheng**, Xinyu Zhang, and Vincent Leung.
  The 20th USENIX Symposium on Networked Systems Design and Implementation **(NSDI)**, 2023.
  (16.0% acceptance rate)
- "UniScatter: a Metamaterial Backscatter Tag for Wideband Joint Communication and Radar Sensing"
  Kun Qian, Lulu Yao, **Kai Zheng**, Xinyu Zhang, and Tse Nga Ng.
  The 29th Annual International Conference On Mobile Computing and Networking **(MobiCom)**, 2023.
  (29.4% acceptance rate)
- "Calibrating a 4-channel fully-digital 60 GHz SDR"
  Aditya Dhananjay, **Kai Zheng**, Jaakko Haarla, Lorenzo Iotti, Marco Mezzavilla, Dennis Shasha, Sundeep Rangan.
  Proceedings of the 14th International Workshop on Wireless Network Testbeds, Experimental Evaluation & Characterization **(WiNTECH)**, 2020.
- "Characterizing 60 GHz patch antenna segments for fully digital transceiver"
  Jaakko Haarla, Vasilii Semkin, **Kai Zheng**, Aditya Dhananjay, Marco Mezzavilla, Juha Ala-Laurinaho, Ville Viikari.
  14th European Conference on Antennas and Propagation **(EuCAP)**, 2020.
- "Software-defined Radios to Accelerate mmWave Wireless Innovation"
  Kai Zheng, Aditya Dhananjay, Marco Mezzavilla, Arjuna Madanayake, Shubhendu Bharadwaj, Viduneth Ariyarathna, Abhimanyu Gosain, et al.
  The IEEE International Symposium on Dynamic Spectrum Access Networks **(DySPAN)**, 2019.
- "MillimeTera: Toward a large-scale open-source mmWave and terahertz experimental testbed"
  Polese, Michele, Francesco Restuccia, Abhimanyu Gosain, Josep Jornet, Shubhendu Bhardwaj, Viduneth Ariyarathna, Soumyajit Mandal, **Kai Zheng**, et al.
  Proceedings of the 3rd ACM Workshop on Millimeter-wave Networks and Sensing Systems **(mmNets)**, 2019.

Journals

- "Real-Time Room Boundary Estimation using Stationary mmWave Radars with Minimal Antennas"
  Kai Zheng, Priyabrata Parida, Vishnu Vardhan Ratnam, Tao Chen, Hao Chen, Hosik Cho, Hyun-Cheol Park.
  IEEE Transactions on Mobile Computing. In revision.
- "NeuroRadar: A Neuromorphic Radar Sensor for Low-Power IoT Systems" (Invited)
  Kai Zheng, Kun Qian, Timothy Woodford, and Xinyu Zhang.
  Communications of the ACM 68.8. 2025.
- "A Neuromorphic Radar Sensor for Low-Power IoT Systems" (Invited, Highlighted)
  Kai Zheng, Kun Qian, Timothy Woodford, and Xinyu Zhang.
  GetMobile: Mobile Computing and Communications. 2024.
- "Structural Pseudocapacitors with Reinforced Interfaces to Increase Multifunctional Efficiency"
  Lulu Yao, **Kai Zheng**, Nandu Koripally, Naresh Eedugurala, Jason D. Azoulay, Xinyu Zhang, Tse Nga Ng.
  Science Advances, 2023.
- "Pi-Radio v1: Calibration techniques to enable fully-digital beamforming at 60 GHz"
  Aditya Dhananjay, **Kai Zheng**, Marco Mezzavilla, Lorenzo Iotti, Dennis Shasha, Sundeep Rangan. Computer Networks. 2021.

Teaching

UC San Diego — **La Jolla, CA**
Teaching Assistant for ECE191 (Group Engineering Project) — *2021–2025*

UC San Diego — **La Jolla, CA**
Teaching Assistant for ECE257A (Modern Communication Networks) — *2024*

Awards

2024: Communications of the ACM Research Highlight

2023: ACM SenSys'23 Best Paper Award (1/179)

Service

Reviewer: IEEE Transactions on Mobile Computing, ACM Transactions on Sensor Networks

Artifacts Reviewer: CoNEXT 2025